

ESAT PRACTICE TEST SERIES

ESAT 2024-25 Recall Mock

Mathematics 1 • Physics • Chemistry

- 27 multiple-choice questions per module
- 40 minutes per module
- Non-calculator
- Hard, recall-informed original practice with full worked solutions

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

ESAT 2024-25 Recall Mock - Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Mathematics 1 answer key

Q1	C	Q2	A	Q3	D	Q4	E	Q5	D	Q6	B	Q7	B	Q8	C	Q9	A
Q10	E	Q11	C	Q12	B	Q13	E	Q14	E	Q15	A	Q16	D	Q17	A	Q18	B
Q19	D	Q20	A	Q21	D	Q22	C	Q23	E	Q24	B	Q25	D	Q26	E	Q27	C

Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

M1-01

Ratio and replacement

Answer **C**

Question 1

A tank contains an alloy in which copper and zinc are in the mass ratio **5 : 3**. Exactly **8 kg** of the well-mixed alloy is removed and replaced by **8 kg** of pure zinc. The new copper-to-zinc ratio is **3 : 2**.

What was the original mass of alloy in the tank?

A 80 kg

B 192 kg

C 200 kgD $\frac{200}{9}$ kg

E 320 kg

WORKED SOLUTION

Let the original mass be M kg. Removing 8 kg of the well-mixed alloy leaves

$$\text{copper} = \frac{5}{8}(M - 8), \quad \text{zinc} = \frac{3}{8}(M - 8).$$

After 8 kg of zinc is added,

$$\frac{\frac{5}{8}(M - 8)}{\frac{3}{8}(M - 8) + 8} = \frac{3}{2}.$$

Hence $10(M - 8) = 9(M - 8) + 192$, so $M = 200$.

Therefore the correct option is **C**.

Trap check: Option B ignores the removed alloy before the zinc is added; option A treats the removal as changing the copper-to-zinc ratio even though a well-mixed sample preserves it; option D reverses the final ratio; option E treats the removed mass as pure copper.

Question 2

For positive integers a and b , the highest common factor of a and b is 6 , and their lowest common multiple is 180 .

How many *ordered* pairs (a, b) satisfy these conditions?

A 8**B** 4**C** 6**D** 3**E** 16**WORKED SOLUTION**

Write $a = 6x$ and $b = 6y$, where x and y are coprime. Then

$$6xy = 180 \implies xy = 30 = 2 \cdot 3 \cdot 5.$$

Each prime factor must be assigned wholly to either x or y . There are $2^3 = 8$ assignments, and order matters.

Therefore the correct option is **A**.

Trap check: Option B counts unordered pairs, C omits the assignments with one reduced factor equal to 1, D merely counts the distinct primes, and E incorrectly includes negative ordered pairs despite the positivity condition.

Question 3

A vehicle travels a route of length D . On one day it covers the first third of the route at speed v and the remainder at speed $2v$, taking **48** minutes in total.

On another day it covers the first half at speed $4v/3$, stops for **6** minutes, and then covers the second half at constant speed kv . The total time is again **48** minutes.

What is k ?

A 2

B $\frac{9}{4}$

C $\frac{5}{2}$

D $\frac{12}{5}$

E $\frac{8}{3}$

WORKED SOLUTION

The first journey gives

$$\frac{D/3}{v} + \frac{2D/3}{2v} = \frac{2D}{3v} = 48,$$

so $D/v = 72$ minutes. On the second day the first half takes

$$\frac{D/2}{4v/3} = \frac{3}{8} \frac{D}{v} = 27 \text{ minutes.}$$

After the six-minute stop, **15** minutes remain. Therefore

$$\frac{D/2}{kv} = \frac{36}{k} = 15, \quad k = \frac{12}{5}.$$

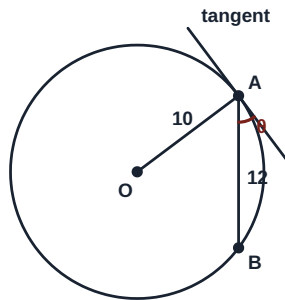
Therefore the correct option is **D**.

Trap check: Do not average segment speeds without checking the associated times; here the arithmetic mean happens to agree because the two segment times are equal.

Question 4

A circle has centre O , radius 10 , and chord AB of length 12 . A tangent is drawn at A . Let θ be the acute angle between the tangent and chord AB .

What is $\tan \theta$?



A $\frac{4}{3}$

B $\frac{3}{5}$

C $\frac{4}{5}$

D $\frac{5}{4}$

E $\frac{3}{4}$

WORKED SOLUTION

Let M be the midpoint of AB . The perpendicular from the centre to a chord bisects it, so $AM = 6$. Hence

$$OM = \sqrt{10^2 - 6^2} = 8.$$

If $\alpha = \angle OAB$, then $\tan \alpha = 8/6 = 4/3$. Since the tangent is perpendicular to OA , $\theta = 90^\circ - \alpha$, and therefore

$$\tan \theta = \frac{1}{\tan \alpha} = \frac{3}{4}.$$

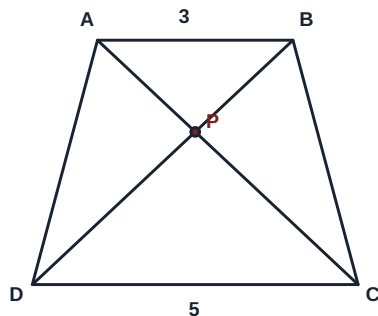
Therefore the correct option is E.

Trap check: The value $4/3$ is $\tan \angle OAB$, not the tangent–chord angle requested.

Question 5

In trapezium $ABCD$, $AB \parallel CD$. The diagonals AC and BD meet at P , and $AB : CD = 3 : 5$. The area of triangle APB is 27.

What is the area of triangle CPD ?



A 45

B 125

C $\frac{243}{25}$ **D 75**

E 192

WORKED SOLUTION

Triangles APB and CPD are similar: they have equal vertically opposite angles at P , and corresponding angles from $AB \parallel CD$. Their linear scale factor is $3 : 5$, so their areas are in the ratio $9 : 25$.

$$27 : \text{area}(CPD) = 9 : 25.$$

Therefore $\text{area}(CPD) = 27 \times 25/9 = 75$.

Therefore the correct option is D.

Trap check: Option A uses the linear scale factor, B cubes it, C reverses the area factor, and E gives the area of the whole trapezium rather than triangle CPD .

Question 6

The sequence begins

$$7, 14, 25, 40, \dots$$

and has constant second differences. Let n be the smallest index for which the n th term exceeds 500, and let e be the amount by which that term exceeds 500.

Which pair (n, e) is correct?

A (15, 31)

B (16, 32)

C (16, 31)

D (15, 32)

E (17, 69)

WORKED SOLUTION

The first differences are 7, 11, 15, ..., so the second difference is 4. Fitting $u_n = 2n^2 + bn + c$ gives

$$u_n = 2n^2 + n + 4.$$

Now $u_{15} = 469$, whereas $u_{16} = 532$. Thus $n = 16$, and the excess is $e = 532 - 500 = 32$.

Therefore the correct option is B.

Trap check: A and D stop one term too early; C subtracts incorrectly after finding the correct index; E moves to the next term after the threshold has already been crossed.

Question 7

A sample has recorded mass **2.400 kg**, correct to the nearest **0.010 kg**, and recorded volume **800 cm³**, correct to the nearest **10 cm³**.

If its density is $\rho \text{ g cm}^{-3}$, which statement gives the exact bounds for ρ ?

A $\frac{479}{159} \leq \rho < \frac{481}{161}$

B $\frac{479}{161} < \rho < \frac{481}{159}$

C $\frac{480}{161} \leq \rho < \frac{480}{159}$

D $\frac{2395}{800} \leq \rho < \frac{2405}{800}$

E $2.995 \leq \rho < 3.005$

WORKED SOLUTION

In grams and cubic centimetres,

$$2395 \leq m < 2405, \quad 795 \leq V < 805.$$

For positive quantities, m/V is smallest as m takes its lower bound and V approaches its excluded upper bound; it is greatest as m approaches its excluded upper bound and V takes its lower bound. Hence

$$\frac{2395}{805} < \rho < \frac{2405}{795}.$$

Dividing numerator and denominator by 5 gives

$$\frac{479}{161} < \rho < \frac{481}{159}.$$

Therefore the correct option is B.

Trap check: The lower endpoint is also excluded: the recorded-volume interval has $V < 805$, so density can approach $2395/805$ but cannot equal it.

Question 8

A bag contains **4** red, **3** blue and **2** green counters. Two counters are drawn at random without replacement.

Given that the counters are different colours, what is the probability that one of them is green?

A $\frac{5}{13}$

B $\frac{6}{13}$

C $\frac{7}{13}$

D $\frac{7}{18}$

E $\frac{14}{27}$

WORKED SOLUTION

Count unordered pairs. The number of different-colour pairs is

$$4 \cdot 3 + 4 \cdot 2 + 3 \cdot 2 = 26.$$

Of these, the pairs involving green number $4 \cdot 2 + 3 \cdot 2 = 14$. Hence the conditional probability is

$$\frac{14}{26} = \frac{7}{13}.$$

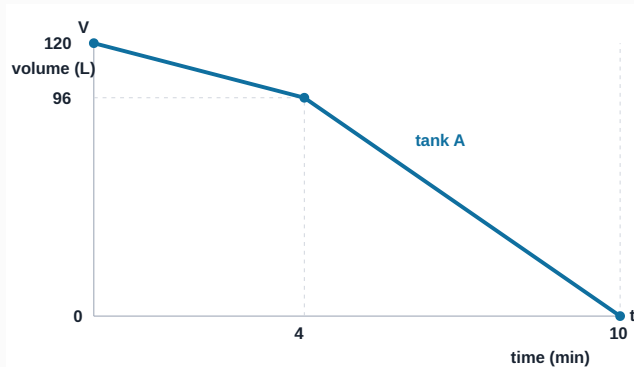
Therefore the correct option is C.

Trap check: The denominator must contain only different-colour pairs, not all $\binom{9}{2}$ pairs.

Question 9

The graph shows the volume V litres of water remaining in tank A, t minutes after draining begins. At $t = 0$, an empty tank B starts filling at a constant rate of **12** litres per minute.

At what time do the two tanks contain the same volume?

**A** $\frac{40}{7}$ min**B** 5 min**C** 6 min**D** $\frac{20}{3}$ min**E** 8 min**WORKED SOLUTION**

At $t = 4$, tank A contains **96** L. From $t = 4$ to $t = 10$, its volume falls by **96** L in **6** min, so the rate is **16** L/min. On this segment,

$$V_A = 96 - 16(t - 4) = 160 - 16t.$$

Tank B contains $V_B = 12t$. Equating volumes gives

$$160 - 16t = 12t \implies t = \frac{40}{7}.$$

This lies between **4** and **10**, so the correct segment was used.

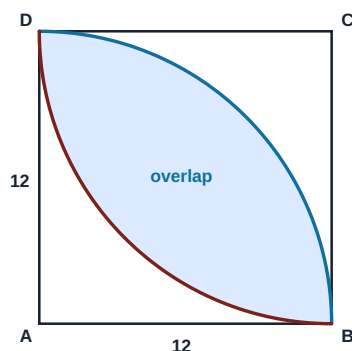
Therefore the correct option is A.

Trap check: The first segment's rule gives a time outside its interval; the change of gradient at 4 minutes matters.

Question 10

A square $ABCD$ has side length **12**. Inside it, quarter-circles of radius **12** are drawn with centres **A** and **C**. The shaded lens is the region lying inside both quarter-circles.

What is the exact area of the shaded region?



A $36\pi - 72$

B 72π

C $144 - 36\pi$

D $72\pi - 72$

E $72\pi - 144$

WORKED SOLUTION

Each arc BD subtends 90° at its centre because adjacent sides of the square are perpendicular. One half of the lens is a 90° sector of radius **12**, less a right triangle with legs **12**:

$$\frac{1}{4}\pi(12)^2 - \frac{1}{2}(12)(12) = 36\pi - 72.$$

The lens has two congruent halves, so its area is

$$2(36\pi - 72) = 72\pi - 144.$$

Therefore the correct option is **E**.

Trap check: A uses only half the lens, B subtracts no triangles, C takes an incorrect square complement, and D subtracts only one of the two right triangles.

Question 11

Let $u = \sqrt{3} + \sqrt{2}$ and $v = \sqrt{3} - \sqrt{2}$.

What is the value of

$$\frac{u^6 + v^6}{u^2 + v^2} ?$$

A 10

B 94

C 97

D 100

E 103

F 970

WORKED SOLUTION

Since $uv = 1$ and $u + v = 2\sqrt{3}$,

$$u^2 + v^2 = (u + v)^2 - 2uv = 10.$$

Using $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$ with $a = u^2$, $b = v^2$,

$$u^6 + v^6 = 10^3 - 3(uv)^2(10) = 1000 - 30 = 970.$$

The required quotient is $970/10 = 97$.

Therefore the correct option is **C**.

Trap check: Option A stops at $u^2 + v^2$, B incorrectly uses $(uv)^2 = 2$, D omits the cubic cross-term, and E changes its sign.

Question 12

The point P lies on the line $y = 2x + 1$ and is equidistant from $A = (0, 4)$ and $B = (6, 0)$. The origin is O .

What is OP^2 ?

A 169

B 218

C 196

D 224

E 49

WORKED SOLUTION

Write $P = (x, 2x + 1)$. Equating squared distances avoids square roots:

$$x^2 + (2x - 3)^2 = (x - 6)^2 + (2x + 1)^2.$$

The quadratic terms cancel, leaving $-12x + 9 = -8x + 37$. Hence $-4x = 28$, so $x = -7$. Therefore $P = (-7, -13)$, and

$$OP^2 = (-7)^2 + (-13)^2 = 49 + 169 = 218.$$

Therefore the correct option is B.

Trap check: A uses only the square of the y -coordinate, C omits x^2 , D introduces a cross-term as if calculating $(x + y)^2$, and E reports only x^2 .

Question 13

Two distinct cards are selected at random from cards numbered **1** to **8**. It is known that their product is even and their sum is **not** divisible by **3**.

What is the probability that their product is divisible by **4**?

A $\frac{9}{22}$

B $\frac{1}{2}$

C $\frac{5}{14}$

D $\frac{7}{11}$

E $\frac{9}{14}$

WORKED SOLUTION

Use unordered pairs. There are **22** pairs with an even product. Of the **10** pairs whose sum is divisible by **3**, two have odd product, so eight are excluded from those **22**. The conditioned sample space therefore has $22 - 8 = 14$ pairs.

The nine admissible pairs with product divisible by **4** are

$$(2, 6), (2, 8), (1, 4), (3, 4), (4, 6), (4, 7), (6, 8), (3, 8), (5, 8).$$

The required probability is **9/14**.

Therefore the correct option is E.

Trap check: A uses the 22 even-product pairs as the denominator and ignores the sum condition; B assumes an unsupported half-and-half split; C counts only a restricted subset centred on the cards 4 and 6; D counts products divisible by 4 before enforcing the sum condition.

Question 14

Let

$$N = 2^4 \cdot 3^3 \cdot 5^2.$$

How many positive divisors of N are divisible by exactly one of **12** and **18**?**A** 9**B** 6**C** 18**D** 33**E** 15**F** 51**WORKED SOLUTION**Divisors divisible by $12 = 2^2 \cdot 3$ number

$$(3)(3)(3) = 27.$$

Divisors divisible by $18 = 2 \cdot 3^2$ number

$$(4)(2)(3) = 24.$$

Those divisible by both are divisible by $\text{lcm}(12, 18) = 36 = 2^2 3^2$, giving

$$(3)(2)(3) = 18.$$

To count divisors in exactly one set, subtract the intersection twice:

$$27 + 24 - 2(18) = 15.$$

Therefore the correct option is E.**Trap check:** Options A and B count divisors in only one named set separately, C counts the intersection, and D is the ordinary inclusion–exclusion total for at least one condition.

Question 15

Containers **A**, **B**, **C** initially hold liquid in the ratio **1 : 3 : 5**. Half of the liquid in **A** is transferred to **B**. Then half of the resulting amount in **B** is transferred to **C**. Finally, **C** contains **100** litres more than **A**.

How much liquid was there originally in total?

A 144 L**B** 200 L**C** 150 L**D** 180 L**E** 108 L**WORKED SOLUTION**

Let the initial amounts be **k**, **3k**, **5k**. After the first transfer,

$$A = \frac{1}{2}k, \quad B = \frac{7}{2}k.$$

Half of this new amount in **B** is $\frac{7k}{4}$, so

$$B = \frac{7}{4}k, \quad C = 5k + \frac{7}{4}k = \frac{27}{4}k.$$

Thus $C - A = (27/4 - 1/2)k = 25k/4 = 100$, giving $k = 16$. The original total was $9k = 144$ L.

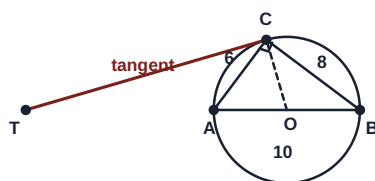
Therefore the correct option is **A**.

Trap check: B ignores the second transfer, C transfers half of B's initial amount, D compares the final amounts in C and B, and E reports C's final amount rather than the original total.

Question 16

Triangle ABC is right-angled at C , with $AC = 6$, $BC = 8$ and $AB = 10$. The circle with diameter AB is shown. The tangent to the circle at C meets the line AB , extended beyond A , at T .

What is CT ?



A $\frac{60}{7}$

B 12

C 15

D $\frac{120}{7}$

E 20

WORKED SOLUTION

Place $A = (0, 0)$ and $B = (10, 0)$. The 6-8-10 triangle gives $C = (18/5, 24/5)$, and the centre is $O = (5, 0)$. The gradient of OC is

$$\frac{24/5}{18/5 - 5} = -\frac{24}{7}.$$

The tangent is perpendicular to OC , so its gradient is $7/24$. Its equation through C is

$$y - \frac{24}{5} = \frac{7}{24} \left(x - \frac{18}{5} \right).$$

Setting $y = 0$ gives $T = (-90/7, 0)$. Therefore

$$CT = \sqrt{\left(\frac{18}{5} + \frac{90}{7} \right)^2 + \left(\frac{24}{5} \right)^2} = \frac{120}{7}.$$

Therefore the correct option is **D**.

Trap check: The tangent is perpendicular to the radius OC , not to either side of triangle ABC .

Question 17

A right-angled triangle has perpendicular sides **9** and **12**. A square is placed with one side on the hypotenuse and the opposite side parallel to it, with the other two vertices lying on the perpendicular sides.

What is the side length of the square?

A $\frac{180}{37}$

B $\frac{36}{7}$

C $\frac{60}{11}$

D $\frac{72}{13}$

E 6

WORKED SOLUTION

The hypotenuse has length **15**, and the altitude from the right angle to the hypotenuse is

$$h = \frac{2 \times (\frac{1}{2})(9)(12)}{15} = \frac{36}{5}.$$

If the square side is s , the small triangle above it is similar to the original triangle. Its scale factor is $(h - s)/h$, so its base, which is a side of the square, satisfies

$$s = 15 \frac{h - s}{h}.$$

Thus $s(h + 15) = 15h$, and with $h = 36/5$,

$$s = \frac{15(36/5)}{15 + 36/5} = \frac{180}{37}.$$

Therefore the correct option is **A**.

Trap check: This is not the standard square placed at the right-angle vertex; here a side of the square lies on the hypotenuse.

Question 18

A data set contains only the values shown.

Value	1	2	3	4	5
Frequency	2	a	4	b	2

There are **20** observations and their mean is **3.1**. Two observations are selected at random without replacement.

What is the probability that they have equal values?

A $\frac{39}{200}$

B $\frac{39}{190}$

C $\frac{31}{190}$

D $\frac{39}{380}$

E $\frac{3}{10}$

WORKED SOLUTION

The total frequency gives $a + b = 12$. The total of all observations is $20(3.1) = 62$, so

$$2 + 2a + 12 + 4b + 10 = 62 \implies a + 2b = 19.$$

Hence $b = 7$ and $a = 5$. The number of unordered equal-value pairs is

$$\binom{2}{2} + \binom{5}{2} + \binom{4}{2} + \binom{7}{2} + \binom{2}{2} = 1 + 10 + 6 + 21 + 1 = 39.$$

There are $\binom{20}{2} = 190$ unordered pairs in total, so the probability is $39/190$.

Therefore the correct option is B.

Trap check: Using 20^2 as the denominator models selection with replacement and also permits selecting the same observation twice.

Question 19

A price is increased by $p\%$, where $p > 0$. The increased price is then reduced by $(p + 10)\%$. The final price is **12%** below the original price.

What is p ?

A -20

B 12

C 22

D 10

E 20

WORKED SOLUTION

Let $x = p/100$. The two multipliers give

$$(1 + x)(0.9 - x) = 0.88.$$

Expanding and simplifying,

$$0.9 - 0.1x - x^2 = 0.88 \implies x^2 + 0.1x - 0.02 = 0.$$

Hence $(x - 0.1)(x + 0.2) = 0$. Since $p > 0$, $x = 0.1$, so $p = 10$.

Therefore the correct option is **D**.

Trap check: A is the rejected algebraic root, B copies the final percentage, C adds 12 and 10, and E neglects the product term in the two successive multipliers.

Question 20

What is the smallest positive integer n for which both $72n$ is a perfect square and $50n$ is a perfect cube?

A 20000**B** 5000**C** 10000**D** 16000**E** 40000**WORKED SOLUTION**

Write $n = 2^a 3^b 5^c$ with the least possible non-negative exponents. Since $72 = 2^3 3^2$, the exponents in $72n$ must be even. Since $50 = 2 \cdot 5^2$, the exponents in $50n$ must be multiples of 3. The least simultaneous choices are

$$a = 5, \quad b = 0, \quad c = 4.$$

Thus $n = 2^5 \cdot 5^4 = 32 \cdot 625 = 20000$.

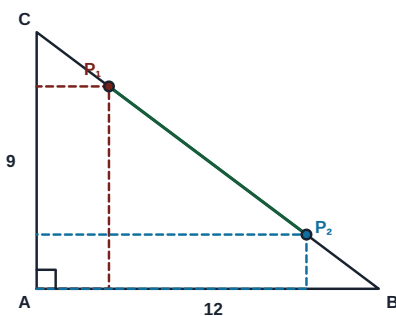
Therefore the correct option is **A**.

Trap check: The same n must satisfy both exponent conditions simultaneously; satisfying only the square condition is insufficient.

Question 21

Triangle ABC is right-angled at A , with $AB = 12$ and $AC = 9$. For a point P on the hypotenuse, perpendiculars from P to AB and AC form a rectangle having A and P as opposite vertices.

There are two positions, P_1 and P_2 , for which the rectangle has area 18. What is P_1P_2 ?



A $3\sqrt{3}$

B $4\sqrt{3}$

C 5

D $5\sqrt{3}$

E $7\sqrt{3}$

WORKED SOLUTION

Place $A = (0, 0)$, $B = (12, 0)$ and $C = (0, 9)$. If $P = (x, y)$, the hypotenuse equation is $y = 9 - \frac{3x}{4}$. The rectangle area condition gives

$$x \left(9 - \frac{3}{4}x \right) = 18 \implies x^2 - 12x + 24 = 0.$$

The two roots are $6 \pm 2\sqrt{3}$, so the horizontal separation of P_1, P_2 is $4\sqrt{3}$. Their vertical separation is $(\frac{3}{4})(4\sqrt{3}) = 3\sqrt{3}$. Hence

$$P_1P_2 = \sqrt{(4\sqrt{3})^2 + (3\sqrt{3})^2} = 5\sqrt{3}.$$

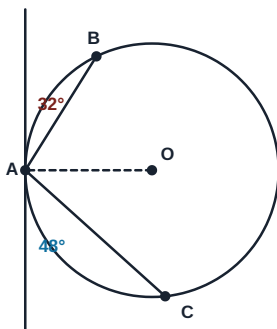
Therefore the correct option is **D**.

Trap check: The horizontal root difference is only $4\sqrt{3}$; P_1P_2 also includes the vertical separation.

Question 22

At a point **A** on a circle, a tangent is drawn. Chords **AB** and **AC** lie on opposite sides of the radius through **A**. The acute angles between the tangent and the chords **AB** and **AC** are 32° and 48° , respectively.

What is the smaller angle **BOC**, where **O** is the centre?



A 100°

B 80°

C 160°

D 200°

E 40°

WORKED SOLUTION

By the tangent–chord theorem,

$$\angle ACB = 32^\circ, \quad \angle ABC = 48^\circ.$$

Therefore $\angle BAC = 180^\circ - 32^\circ - 48^\circ = 100^\circ$. The arc **BC** not containing **A** has central angle 200° . The smaller angle at the centre is therefore

$$360^\circ - 200^\circ = 160^\circ.$$

Therefore the correct option is **C**.

Trap check: A gives the inscribed angle, B adds the two tangent–chord angles without doubling, D selects the reflex central angle, and E doubles their difference.

Question 23

An increasing arithmetic sequence has positive integer terms. Its terms satisfy

$$u_6^2 = u_3 u_{11} \quad \text{and} \quad u_{11} - u_3 = 48.$$

What is the first term of the sequence?

A 6

B 9

C 12

D 18

E 15

WORKED SOLUTION

Let the first term be a and common difference be $d > 0$. Then

$$u_3 = a + 2d, \quad u_6 = a + 5d, \quad u_{11} = a + 10d.$$

The difference condition gives $8d = 48$, so $d = 6$. Expanding the other condition,

$$(a + 5d)^2 = (a + 2d)(a + 10d)$$

$$a^2 + 10ad + 25d^2 = a^2 + 12ad + 20d^2,$$

so $5d^2 = 2ad$. Since $d > 0$, $a = 5d/2 = 15$.

Therefore the correct option is **E**.

Trap check: The identity is multiplicative, not the ordinary arithmetic-mean relation for u_3 , u_6 and u_{11} .

Question 24

Two fair six-sided dice are thrown.

Given that the product of the two scores is divisible by 6, what is the probability that their sum is prime?

A $\frac{7}{15}$

B $\frac{8}{15}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{9}$

WORKED SOLUTION

The 15 ordered outcomes whose product is divisible by 6 are

(1, 6); (2, 3), (2, 6); (3, 2), (3, 4), (3, 6); (4, 3), (4, 6); (5, 6); (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6).

Of these, the eight outcomes

(1, 6), (6, 1), (2, 3), (3, 2), (3, 4), (4, 3), (5, 6), (6, 5)

have prime sum. The required probability is therefore $\frac{8}{15}$.

Therefore the correct option is B.

Trap check: A misses one favourable ordered outcome, C counts unordered pairs, D includes one non-prime sum, and E uses the eight favourable outcomes over the original 36-outcome sample space.

Question 25

The arithmetic sequence

$$5, 12, 19, 26, \dots$$

contains terms that are divisible by **11** but not divisible by **3**. What is the position of the **15th** such term?

A 214

B 225

C 236

D 247

E 258

F 159

WORKED SOLUTION

The n th term is $7n - 2$. Divisibility by **11** requires

$$7n \equiv 2 \pmod{11},$$

so $n \equiv 5 \pmod{11}$. Write $n = 5 + 11k$, where $k \geq 0$. The term is divisible by **3** precisely when $n \equiv 2 \pmod{3}$. Since

$$5 + 11k \equiv 2 + 2k \pmod{3},$$

this happens when $2k \equiv 0 \pmod{3}$, so $k \equiv 0 \pmod{3}$. These values must be excluded. The allowed values of k are therefore

$$1, 2, 4, 5, 7, 8, \dots$$

There are two allowed values in each block of three. The **15th** allowed value is $k = 22$, giving

$$n = 5 + 11(22) = 247.$$

Indeed, $7(247) - 2 = 1727 = 11 \times 157$, which is not divisible by **3**.

Therefore the correct option is D.

Trap check: C excludes the wrong residue class of k ; F counts the 15th multiple-of-11 candidate without any exclusion; A, B and E are one-block or one-candidate counting shifts.

Question 26

A positive real number x satisfies

$$x + \frac{1}{x} = 3.$$

For positive integers n , define

$$A_n = x^n + \frac{1}{x^n}.$$

What is the remainder when A_{2025} is divided by 10?

A 2

B 3

C 5

D 7

E 8

WORKED SOLUTION

Multiplying A_n by $x + x^{-1} = 3$ gives

$$3A_n = A_{n+1} + A_{n-1},$$

so $A_{n+1} = 3A_n - A_{n-1}$. Also $A_0 = 2$ and $A_1 = 3$. Working modulo 10,

$$A_0, A_1, A_2, A_3, A_4, A_5, A_6 \equiv 2, 3, 7, 8, 7, 3, 2.$$

The pair $(A_6, A_7) \equiv (2, 3)$ repeats the initial pair, so the remainders have period 6. Since 2025 leaves remainder 3 on division by 6,

$$A_{2025} \equiv A_3 \equiv 8 \pmod{10}.$$

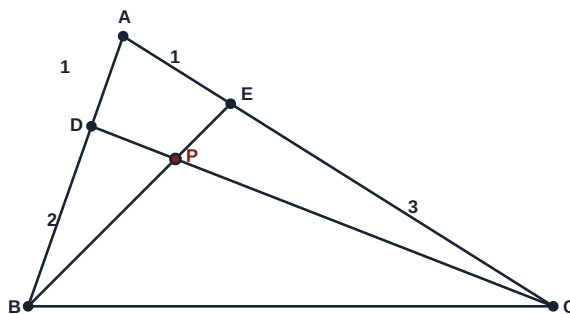
Therefore the correct option is E.

Trap check: Track a pair of consecutive remainders to justify the cycle; matching one term alone is not enough.

Question 27

In triangle ABC , point D lies on AB with $AD : DB = 1 : 2$, and point E lies on AC with $AE : EC = 1 : 3$. Lines DC and EB intersect at P .

What is $EP : PB$?



A 3 : 7

B 8 : 3

C 3 : 8

D 4 : 9

E 2 : 7

F 3 : 11

G 11 : 3

H 8 : 11

WORKED SOLUTION

Choose convenient coordinates $A = (0, 0)$, $B = (3, 0)$, $C = (0, 4)$. Then $D = (1, 0)$ and $E = (0, 1)$. Parameterise P on both cevians:

$$P = D + t(C - D) = (1 - t, 4t),$$

$$P = E + s(B - E) = (3s, 1 - s).$$

Solving $1 - t = 3s$ and $4t = 1 - s$ gives $s = 3/11$. Since s is the fraction of the way from E to B ,

$$EP : PB = s : (1 - s) = \frac{3}{11} : \frac{8}{11} = 3 : 8.$$

Therefore the correct option is **C**.

Trap check: B reverses the required ratio; F reports $EP : EB = 3 : 11$; G reverses that whole-segment ratio; H reports $PB : EB = 8 : 11$. A, D and E arise from using one of the side ratios directly or mismatching coefficients in the two line parameters.

ANSWER KEY

ESAT 2024-25 Recall Mock - Physics

27 hard, recall-informed Physics questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Physics answer key

Q1	E	Q2	B	Q3	A	Q4	C	Q5	E	Q6	B	Q7	C	Q8	A	Q9	B
Q10	D	Q11	C	Q12	E	Q13	C	Q14	C	Q15	A	Q16	C	Q17	E	Q18	B
Q19	B	Q20	C	Q21	D	Q22	E	Q23	C	Q24	B	Q25	D	Q26	A	Q27	E

Physics

27 hard, recall-informed Physics questions with complete worked solutions.

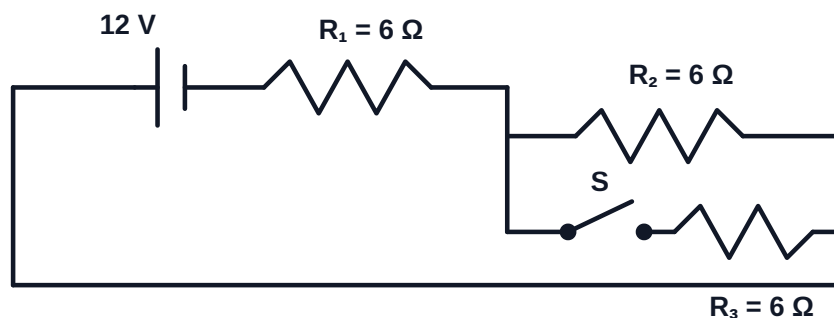
P-01

Qualitative circuit changes

Answer **E**

Question 1

Three identical $6\ \Omega$ resistors are connected to an ideal $12\ \text{V}$ source as shown. Initially switch S is open; it is then closed.



Which statements about closing S are correct?

1. The current supplied by the source increases.
2. The potential difference across R_1 increases.
3. The power dissipated in R_2 decreases.

A 1 only**B** 2 only**C** 1 and 3 only**D** 2 and 3 only**E** 1, 2 and 3

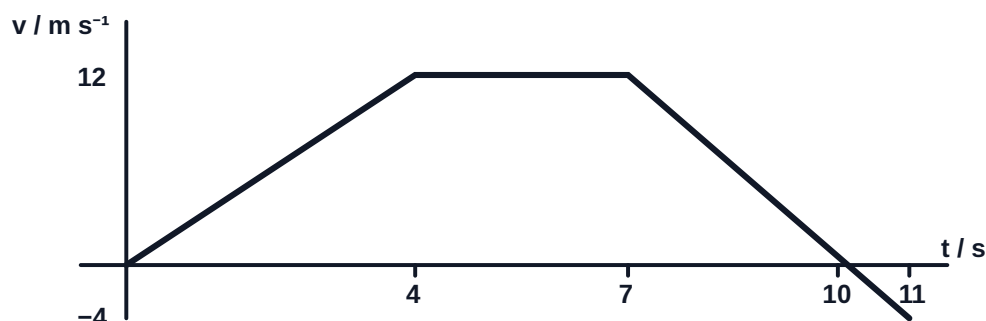
WORKED SOLUTION

With S open, the total resistance is $12\ \Omega$, so the source current is $1\ \text{A}$. The voltage across R_1 is $6\ \text{V}$, and R_2 dissipates $6^2/6 = 6\ \text{W}$.

Closing S puts R_2 and R_3 in parallel: their resistance is $3\ \Omega$, so the total is $9\ \Omega$. The source current rises to $4/3\ \text{A}$, the voltage across R_1 rises to $8\ \text{V}$, and the parallel branch voltage becomes $4\ \text{V}$. Thus R_2 's power falls to $4^2/6 = 8/3\ \text{W}$. All three statements are correct, so the answer is **E**.

Question 2

A particle starts from rest. It accelerates uniformly to 12 m s^{-1} in 4 s , travels at this velocity for 3 s , then changes velocity uniformly to -4 m s^{-1} over the next 4 s . What are its total displacement and total distance travelled?



A 80 m, 80 m

B 76 m, 80 m

C 80 m, 84 m

D 76 m, 76 m

E 72 m, 76 m

WORKED SOLUTION

Signed area gives displacement:

$$\frac{1}{2}(4)(12) + (3)(12) + \frac{1}{2}(12-4)(4) = 24 + 36 + 16 = 76 \text{ m.}$$

In the final section the velocity reaches zero after 3 s . Its distance in that section is $\frac{1}{2}(3)(12) + \frac{1}{2}(1)(4) = 20 \text{ m}$. Thus total distance is $24 + 36 + 20 = 80 \text{ m}$. The correct option is **B**.

Question 3

Five complete cycles of a wave pass a fixed point in medium X in 0.020 s . Adjacent crests in X are 0.80 m apart. The wave then enters medium Y obliquely, where its speed is $\frac{3}{4}$ of its speed in X . Which row gives the speed, frequency and wavelength in Y , together with the direction of bending?

A 150 m s^{-1} , 250 Hz , 0.60 m ; towards the normal

B 150 m s^{-1} , 187.5 Hz , 0.80 m ; towards the normal

C 200 m s^{-1} , 250 Hz , 0.60 m ; away from the normal

D 150 m s^{-1} , 250 Hz , 0.60 m ; away from the normal

E 200 m s^{-1} , 250 Hz , 0.80 m ; no change in direction

WORKED SOLUTION

The frequency is $f = 5/0.020 = 250\text{ Hz}$. In X , $v_X = f\lambda_X = 250(0.80) = 200\text{ m s}^{-1}$. Therefore

$$v_Y = \frac{3}{4}v_X = 150\text{ m s}^{-1}, \quad \lambda_Y = \frac{v_Y}{f} = \frac{150}{250} = 0.60\text{ m}.$$

Frequency does not change at the boundary. Since the wave slows down, it bends towards the normal. The correct option is **A**.

Question 4

Two rods P and Q of equal cross-sectional area are joined end to end. Rod P has length L and thermal conductivity k . Rod Q has length $2L$ and thermal conductivity $3k$. The free end of P is kept at 100°C , and the free end of Q at 20°C . Assume one-dimensional conduction: the rods' sides are thermally insulated, contact resistance at the junction is negligible, and no heat is lost to the surroundings. In steady state, the transfer rate through a rod is $kA\Delta T/L$. What is the junction temperature?

A 36°C

B 44°C

C 52°C

D 60°C

E 68°C

F 48°C

WORKED SOLUTION

The thermal resistances are proportional to $L/(kA)$. Thus

$$R_P : R_Q = 1 : \frac{2}{3} = 3 : 2.$$

The total temperature drop is 80°C , so the drop across P is $\frac{3}{5}(80) = 48^\circ\text{C}$. The junction is therefore at $100 - 48 = 52^\circ\text{C}$. The correct option is **C**.

Question 5

A simple AC generator contains a coil of N turns rotating at angular speed ω in a uniform magnetic field and is connected to a fixed resistor. For this generator, $V_{\text{peak}} \propto N\omega$, $f \propto \omega$, and the average power in the resistor is proportional to V_{peak}^2 .

The coil is replaced by one of $2N$ turns rotating at angular speed $\omega/2$, with all other factors unchanged. How do the peak voltage and the energy delivered to the resistor in one complete cycle change?

A Peak voltage doubles; energy per cycle doubles

B Peak voltage is unchanged; energy per cycle is unchanged

C Peak voltage halves; energy per cycle halves

D Peak voltage doubles; energy per cycle is four times as large

E Peak voltage is unchanged; energy per cycle doubles

WORKED SOLUTION

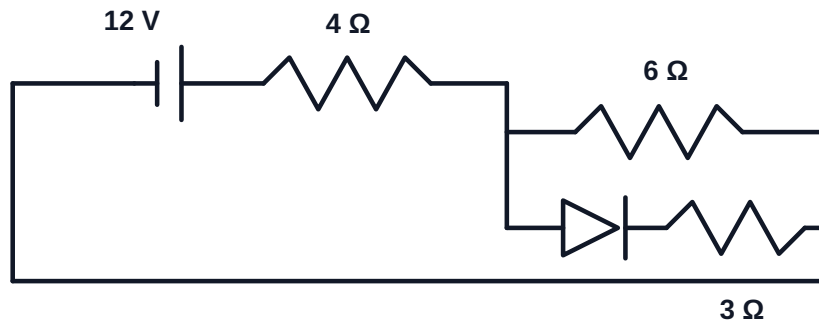
The peak induced voltage scales as $N\omega$, so the two changes multiply it by $2 \times \frac{1}{2} = 1$: the peak voltage is unchanged. The average power in the fixed resistor is therefore unchanged.

The frequency halves, so the period of one cycle doubles. Since energy per cycle equals average power multiplied by the period, the energy delivered in each complete cycle doubles. The correct option is **E**.

Question 6

An ideal **12 V** source supplies a **4 Ω** resistor in series with two parallel branches. One branch is **6 Ω**; the other contains an ideal diode in series with **3 Ω**. With the first polarity the diode conducts. When the source polarity is reversed, the diode blocks. What is the ratio

$$\frac{\text{power drawn with the diode conducting}}{\text{power drawn with the diode blocking}}?$$



A $\frac{2}{3}$

B $\frac{5}{3}$

C $\frac{3}{2}$

D $\frac{5}{2}$

E $\frac{10}{3}$

WORKED SOLUTION

When the diode conducts, **6 Ω** and **3 Ω** are in parallel, giving **2 Ω**. Total resistance is **6 Ω**, so $P_{\text{on}} = V^2/R = 144/6 = 24 \text{ W}$.

When it blocks, total resistance is **4 + 6 = 10 Ω**, so $P_{\text{off}} = 144/10 = 14.4 \text{ W}$. Therefore $P_{\text{on}}/P_{\text{off}} = 24/14.4 = 5/3$. The correct option is **B**.

Question 7

A Geiger counter records the following counts in repeated **30 s** intervals. The source and detector remain fixed.

Measurement	1	2	3	4
Source + background	148	161	153	139
Background only	18	22	16	24

Which statements are correct?

1. Repeated counts can differ even if the source activity and apparatus are unchanged.
2. The best estimate from these data of the source-only count in **30 s** is **130.25**.
3. Subtracting the mean background removes all random uncertainty from the source estimate.

A 1 only

B 2 only

C 1 and 2 only

D 2 and 3 only

E 1, 2 and 3

WORKED SOLUTION

Radioactive decay is random, so statement 1 is correct. The mean source-plus-background count is $(148 + 161 + 153 + 139)/4 = 150.25$, while the mean background is $(18 + 22 + 16 + 24)/4 = 20$. Their difference is **130.25**, so statement 2 is correct.

Background subtraction corrects the central estimate but does not eliminate fluctuations in either set of counts, so statement 3 is false. The correct option is **C**.

Question 8

A **1200 kg** car pulls an **800 kg** trailer along a horizontal road. The resistive forces on the car and trailer are **400 N** and **200 N**, respectively. Both accelerate forwards at **1.5 m s^{-2}** . What are the driving force on the car and the force exerted by the trailer on the car?

A 3600 N forwards; 1400 N backwards

B 3000 N forwards; 1200 N backwards

C 3600 N forwards; 1200 N backwards

D 3400 N forwards; 1400 N backwards

E 3600 N forwards; 1400 N forwards

WORKED SOLUTION

For the complete car-trailer system, if the driving force is F , then

$$F - (400 + 200) = (1200 + 800)(1.5),$$

so $F = 3600 \text{ N}$. For the trailer alone, if the coupling force is T ,

$$T - 200 = 800(1.5),$$

giving $T = 1400 \text{ N}$. By Newton's third law, the trailer exerts an equal **1400 N** force backwards on the car. The correct option is **A**.

Question 9

A sound source emits **240 Hz** while moving directly towards a stationary observer at **60 m s^{-1}** . The speed of sound in the still air is **360 m s^{-1}** . During one source period, a crest travels at the speed of sound while the source moves closer to the preceding crest. What frequency does the observer detect?

A 270 Hz

B 288 Hz

C 240 Hz

D 300 Hz

E 320 Hz

WORKED SOLUTION

One source period is $T = 1/240 \text{ s}$. In this time the earlier crest travels $360T$, while the source advances $60T$. The wavelength in front of the source is therefore

$$\lambda = (360 - 60)T = \frac{300}{240} = 1.25 \text{ m}.$$

The stationary observer receives waves travelling at **360 m s^{-1}** , so

$$f_{\text{obs}} = \frac{360}{1.25} = 288 \text{ Hz}.$$

The correct option is **B**.

Question 10

A transformer has **240 V** across its primary and **12 V** across its secondary. A **6 Ω** resistor is connected across the secondary. The transformer is 80% efficient, where efficiency is secondary output power divided by primary input power. Assume that its terminal voltage ratio still obeys $V_s/V_p = N_s/N_p$. What are the primary current I_p and the turns ratio N_s/N_p ?

A $I_p = \frac{1}{10} \text{ A}, N_s/N_p = \frac{1}{20}$

B $I_p = \frac{1}{8} \text{ A}, N_s/N_p = \frac{1}{10}$

C $I_p = \frac{1}{10} \text{ A}, N_s/N_p = \frac{1}{10}$

D $I_p = \frac{1}{8} \text{ A}, N_s/N_p = \frac{1}{20}$

E $I_p = \frac{1}{5} \text{ A}, N_s/N_p = \frac{1}{20}$

WORKED SOLUTION

The secondary current is $12/6 = 2 \text{ A}$, so output power is $12 \times 2 = 24 \text{ W}$. Input power is $24/0.80 = 30 \text{ W}$, giving $I_p = 30/240 = 1/8 \text{ A}$. Also

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{12}{240} = \frac{1}{20}.$$

The correct option is **D**.

Question 11

A fixed mass of ideal gas initially has pressure p , volume V , absolute temperature T and density ρ . It is first compressed isothermally to volume $V/2$. Keeping this new volume fixed, it is then heated until its absolute temperature is $3T/2$. What are its final pressure and density? For this question, use $pV/T = \text{constant}$ for a fixed mass of ideal gas.

A $\frac{3}{2}p, 2\rho$

B $2p, \frac{3}{2}\rho$

C $3p, 2\rho$

D $3p, 3\rho$

E $4p, 2\rho$

WORKED SOLUTION

During the isothermal compression, pV is constant. Halving the volume therefore doubles the pressure to $2p$. At fixed volume, pressure is proportional to absolute temperature, so the second step multiplies the pressure by $3/2$:

$$p_{\text{final}} = 2p \left(\frac{3}{2} \right) = 3p.$$

The mass is unchanged and the final volume is $V/2$, so $\rho_{\text{final}} = m/(V/2) = 2\rho$. The correct option is **C**.

Question 12

An object falls vertically through a fluid. Its weight is W , its upthrust is a constant U , and its drag force is kv^2 , where v is its speed. Its terminal speed is v_t . What is the resultant force when its speed is $v_t/2$?

A $\frac{1}{4}(W - U)$ downward

B $\frac{1}{2}(W - U)$ downward

C $\frac{3}{4}(W - U)$ upward

D $\frac{1}{4}(W - U)$ upward

E $\frac{3}{4}(W - U)$ downward

WORKED SOLUTION

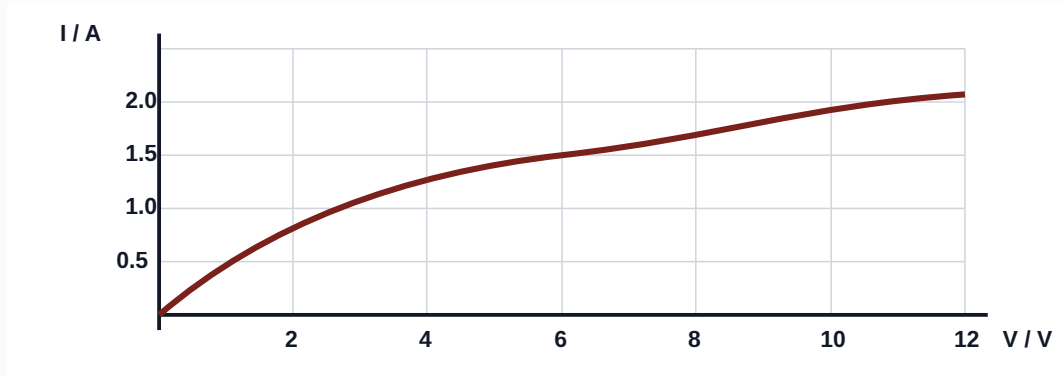
At terminal speed, $kv_t^2 = W - U$. At $v_t/2$, the drag is $\frac{1}{4}kv_t^2 = \frac{1}{4}(W - U)$. The downward resultant is therefore

$$(W - U) - \frac{1}{4}(W - U) = \frac{3}{4}(W - U).$$

The correct option is **E**.

Question 13

The graph shows the current-voltage characteristic of a filament lamp. The lamp is connected in series with a 4Ω resistor across an ideal 12 V source. What power is dissipated in the lamp?

**A** 4.5 W**B** 6 W**C** 9 W**D** 12 W**E** 18 W**WORKED SOLUTION**

If the lamp voltage is V_L and the current is I , the series resistor has voltage $4I$, so

$$V_L + 4I = 12.$$

From the graph, the lamp has $I = 1.5\text{ A}$ at $V_L = 6\text{ V}$; this satisfies $6 + 4(1.5) = 12$. The increasing lamp curve and decreasing load line give a unique operating point. Thus $P_L = V_L I = 6(1.5) = 9\text{ W}$. The correct option is **C**.

Question 14

Equal masses of solids **A** and **B**, both initially at **20°C**, are heated by identical constant-power heaters. Energy loss is negligible. The results are:

Solid	Melting temperature	Time to reach melting temperature	Time at constant temperature while melting
A	60°C	40 s	50 s
B	80°C	60 s	30 s

Which statements are correct?

1. Solid B has the higher melting point.
2. In the solid state, A has the greater specific heat capacity.
3. A has the greater specific latent heat of fusion.

A 1 only

B 2 only

C 1 and 3 only

D 2 and 3 only

E 1, 2 and 3

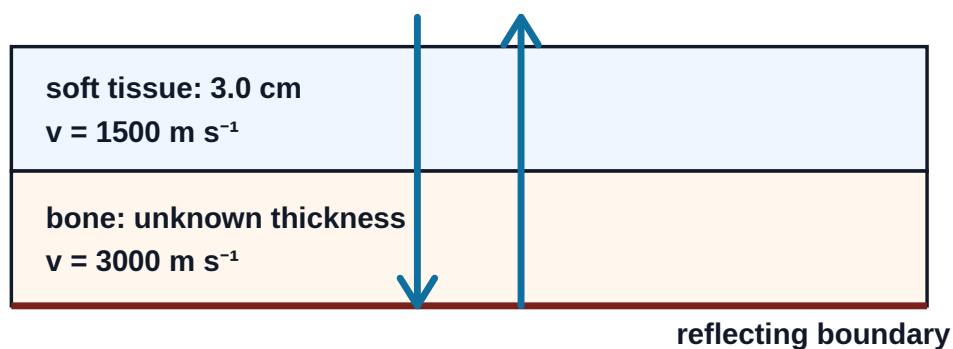
WORKED SOLUTION

The melting temperatures make statement 1 true. Before melting, $Pt = mc\Delta T$. For A, $t/\Delta T = 40/40 = 1 \text{ s K}^{-1}$; for B, $60/60 = 1 \text{ s K}^{-1}$. The masses and powers are equal, so the specific heat capacities are equal and statement 2 is false.

During melting, $Pt = mL$. A remains on the plateau for longer, so A has the greater latent heat and statement 3 is true. The correct option is **C**.

Question 15

An ultrasound pulse travels normally through **3.0 cm** of soft tissue at **1500 m s⁻¹**, then through bone at **3000 m s⁻¹**. It reflects from a boundary within the bone and returns along the same path. The echo is received **60 μs** after emission. How far below the surface is the reflecting boundary?

**A** 6.0 cm**B** 3.0 cm**C** 4.5 cm**D** 7.5 cm**E** 9.0 cm**WORKED SOLUTION**

The round-trip time in the soft tissue is

$$t_s = \frac{2(0.030)}{1500} = 40 \mu\text{s}.$$

The remaining **20 μs** is the round-trip time in bone. If the one-way bone thickness is **x**,

$$\frac{2x}{3000} = 20 \times 10^{-6} \Rightarrow x = 0.030 \text{ m} = 3.0 \text{ cm}.$$

The reflecting boundary is therefore **3.0 + 3.0 = 6.0 cm** below the surface. The correct option is **A**.

Question 16

An electric pump takes in power at **500 W** and is **80%** efficient. It pumps water at **2.0 kg s⁻¹** from a large reservoir to an outlet **15 m** higher. The reservoir surface and outlet are both at atmospheric pressure, and the speed of the reservoir surface is negligible. Neglect all losses other than those represented by the efficiency, and take **$g = 10 \text{ N kg}^{-1}$** .

What is the speed of the water as it leaves the outlet?

A 5 m s^{-1}

B $5\sqrt{2} \text{ m s}^{-1}$

C 10 m s^{-1}

D $10\sqrt{2} \text{ m s}^{-1}$

E 20 m s^{-1}

WORKED SOLUTION

The useful power is $0.80(500) = 400 \text{ W}$. Raising **2.0 kg** each second through **15 m** requires gravitational power

$$\dot{m}gh = 2(10)(15) = 300 \text{ W}.$$

The remaining **100 W** becomes kinetic energy of the water each second:

$$\frac{1}{2}\dot{m}v^2 = 100 \Rightarrow \frac{1}{2}(2)v^2 = 100.$$

Thus **$v = 10 \text{ m s}^{-1}$** . The correct option is **C**.

Question 17

The north pole of a bar magnet moves directly towards one end of a coil at speed **u** , is stopped, and is then pulled away. As it passes the same positions on the way out, its speed is **$u/2$** . Which statements are correct?

1. While the north pole approaches, the near face of the coil behaves as a north pole.
2. While the magnet is stationary, the induced potential difference is zero.
3. While the magnet recedes through the same positions, the induced potential difference has the opposite direction and a smaller magnitude than during approach.

A 1 only

B 2 only

C 1 and 2 only

D 2 and 3 only

E 1, 2 and 3

WORKED SOLUTION

Lenz's law makes the near face repel the approaching north pole, so it acts north. No flux change occurs while the magnet is stationary. On withdrawal, the flux change reverses, reversing the induced voltage; at the same position its rate of change is smaller because the speed is halved. All statements are correct, so the answer is **E**.

Question 18

A source emits alpha, beta-minus and gamma radiation. A detector has a background rate of **20 counts min⁻¹**. It records:

- **620 counts min⁻¹** with no absorber;
- **420 counts min⁻¹** after paper is inserted;
- **170 counts min⁻¹** after aluminium is added behind the paper;
- **20 counts min⁻¹** after thick lead is added.

Assume the paper removes only alpha, the aluminium then removes only beta-minus, and the lead removes the remaining gamma. Which option gives, respectively, the ratio of the source's beta-minus count rate to its gamma count rate and the fraction of the source-only count rate due to alpha radiation?

A 2 : 1; $\frac{1}{3}$

B 5 : 3; $\frac{1}{3}$

C 3 : 2; $\frac{10}{31}$

D 5 : 4; $\frac{1}{2}$

E 3 : 5; $\frac{1}{3}$

WORKED SOLUTION

The beta-minus component is the decrease when aluminium is added:

$$420 - 170 = 250 \text{ counts min}^{-1}.$$

After paper and aluminium, the source contribution is the gamma rate. Subtracting background gives

$$170 - 20 = 150 \text{ counts min}^{-1}.$$

The alpha component is the decrease when paper is inserted, $620 - 420 = 200 \text{ counts min}^{-1}$. The total source-only rate is $620 - 20 = 600 \text{ counts min}^{-1}$, so the alpha fraction is $200/600 = 1/3$.

Thus the required pair is **5 : 3; 1/3**. The correct option is **B**.

Question 19

A vertical-sided container of base area A holds an oil layer of density ρ and depth $2h$ above a liquid of density 2ρ and depth h . The same volumes of both liquids are transferred to a second vertical-sided container of base area $2A$. Compared with the first container, how do the gauge pressure and the total force due to liquid pressure on the horizontal base change?

- A** Pressure halves; force halves
- B** Pressure halves; force is unchanged
- C** Pressure is unchanged; force doubles
- D** Pressure becomes one quarter; force halves
- E** Pressure doubles; force is unchanged

WORKED SOLUTION

Initially, $p_1 = \rho g(2h) + 2\rho g(h) = 4\rho gh$. Doubling base area halves each depth, so

$$p_2 = \rho g(h) + 2\rho g\left(\frac{h}{2}\right) = 2\rho gh = \frac{p_1}{2}.$$

But $F_1 = p_1 A = 4\rho ghA$, while $F_2 = p_2(2A) = 4\rho ghA$. Pressure halves and force is unchanged. The correct option is **B**.

Question 20

A positively charged rod is brought close to an initially neutral isolated metal sphere without touching it. While the rod remains nearby, the sphere is briefly connected to Earth. The Earth connection is removed before the rod is taken away. What is the final charge on the sphere, and what happened while it was earthed?

- A** Neutral; equal numbers of electrons flowed in both directions
- B** Positive; electrons flowed from Earth to the sphere
- C** Negative; electrons flowed from Earth to the sphere
- D** Positive; electrons flowed from the sphere to Earth
- E** Negative; positive charge flowed from Earth to the sphere

WORKED SOLUTION

The positive rod attracts electrons towards the sphere. While earthed, additional electrons flow from Earth onto it. Removing the Earth connection traps this excess; after the rod is removed, the electrons redistribute and the sphere remains negative. The correct option is **C**.

Question 21

Two otherwise identical hot metal containers are at the same temperature in identical cooler surroundings. The initial net radiative loss from the dull black container is four times that from the polished silver container. Each also loses energy by non-radiative processes at the same rate H . The measured initial total loss rates are **70 W** for the black container and **40 W** for the silver container. What fraction of the black container's total loss is by radiation?

A $\frac{3}{7}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{4}{7}$

E $\frac{5}{7}$

WORKED SOLUTION

Let the black container's radiative loss be R . The silver container's radiative loss is then $R/4$. Hence

$$R + H = 70, \quad \frac{R}{4} + H = 40.$$

Subtracting gives $3R/4 = 30$, so $R = 40 \text{ W}$. The required fraction is therefore

$$\frac{R}{70} = \frac{40}{70} = \frac{4}{7}.$$

The correct option is **D**.

Question 22

A particle of mass m travels at speed $3u$. It collides with a particle of mass $2m$ travelling in the same direction at speed u . They stick together. Which option gives their common speed and the fraction of the initial kinetic energy that is lost?

A $\frac{5u}{3}; \frac{8}{25}$

B $\frac{4u}{3}; \frac{1}{4}$

C $\frac{5u}{3}; \frac{4}{33}$

D $\frac{4u}{3}; \frac{1}{3}$

E $\frac{5u}{3}; \frac{8}{33}$

WORKED SOLUTION

Momentum conservation gives $3mu + 2mu = 3mv$, so $v = 5u/3$. Initial kinetic energy is $\frac{1}{2}m(3u)^2 + \frac{1}{2}(2m)u^2 = \frac{11}{2}mu^2$. Final kinetic energy is

$$\frac{1}{2}(3m)\left(\frac{5u}{3}\right)^2 = \frac{25}{6}mu^2.$$

The loss is $\frac{4}{3}mu^2$, so the fraction lost is $(4/3)/(11/2) = 8/33$. The required pair is therefore **5u/3; 8/33**, and the correct option is **E**.

Question 23

In medium X , a wave has speed v , wavelength λ and period T . It enters normally into a slab of medium Y , where its speed is $3v/4$. The slab thickness is 6λ . How many wavelengths fit across the slab, and how long does the wave take to cross it?

A 6 wavelengths; $6T$

B 8 wavelengths; $6T$

C 8 wavelengths; $8T$

D 6 wavelengths; $8T$

E 9 wavelengths; $8T$

WORKED SOLUTION

Frequency is unchanged, so the wavelength in Y is $\lambda_Y = (3/4)\lambda$. Therefore

$$\frac{6\lambda}{(3/4)\lambda} = 8$$

wavelengths fit in the slab. Crossing time is $6\lambda/(3v/4) = 8\lambda/v = 8T$, because $T = \lambda/v$. The correct option is **C**.

Question 24

In an original transmission system, power P enters cables of total resistance R at voltage V . The power lost in the cables is exactly $P/5$.

In a redesigned system, the same generator power P enters a transformer of efficiency **80%**, whose output sends power through the same cables at voltage $4V$. At the far end, a second transformer of efficiency **75%** supplies a load.

What fraction of P reaches the load?

A $\frac{3}{5}$

B $\frac{297}{500}$

C $\frac{99}{125}$

D $\frac{12}{25}$

E $\frac{99}{100}$

WORKED SOLUTION

Originally $I = P/V$ and $I^2 R = P/5$. The first transformer sends $4P/5$ into the cables at $4V$, so the new cable current is

$$I' = \frac{4P/5}{4V} = \frac{P}{5V} = \frac{I}{5}.$$

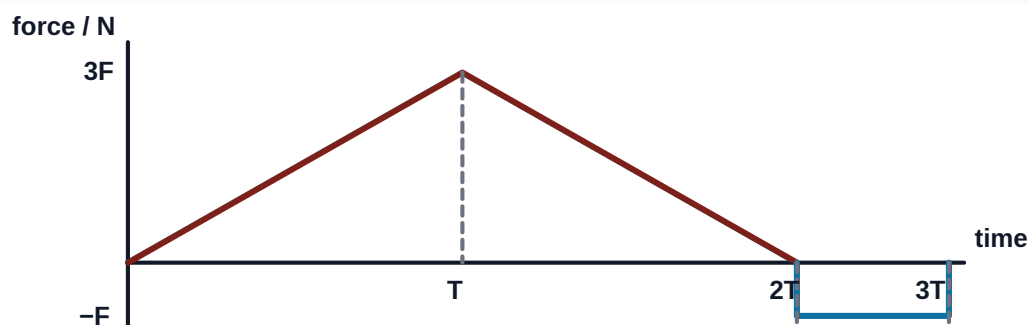
The new cable loss is therefore $(I/5)^2 R = (P/5)/25 = P/125$. The power entering the receiving transformer is

$$\frac{4P}{5} - \frac{P}{125} = \frac{99P}{125}.$$

Its **75%** efficiency gives load power $(3/4)(99P/125) = 297P/500$. The correct option is **B**.

Question 25

A trolley of mass m is initially at rest. The horizontal force on it varies with time as shown. Immediately after $t = 3T$, it collides with a stationary trolley of mass $3m$; the trolleys stick together. They then move up a smooth ramp and come momentarily to rest at vertical height h .



Which expression gives h ?

A $\frac{F^2 T^2}{2m^2 g}$

B $\frac{F^2 T^2}{4m^2 g}$

C $\frac{F^2 T^2}{32m^2 g}$

D $\frac{F^2 T^2}{8m^2 g}$

E $\frac{9F^2 T^2}{32m^2 g}$

WORKED SOLUTION

The positive triangular area is $\frac{1}{2}(2T)(3F) = 3FT$; the negative rectangular area is $-FT$. The net impulse is therefore $2FT$, so the first trolley's momentum just before collision is $2FT$.

Momentum conservation in the sticking collision gives the combined mass $4m$ speed

$$v = \frac{2FT}{4m} = \frac{FT}{2m}.$$

On the smooth ramp, $\frac{1}{2}(4m)v^2 = (4m)gh$, so $h = v^2/(2g) = F^2 T^2/(8m^2 g)$. The correct option is **D**.

Question 26

A lift of mass m is moving vertically upward. At three instants X , Y and Z , its upward speed has the same value v . Its acceleration is upward with magnitude a at X , zero at Y , and downward with magnitude a at Z , where $0 < a < g$. The cable remains taut and resistive forces are negligible.

Which ordering is correct for the instantaneous powers P_X, P_Y, P_Z delivered to the lift by the cable?

A $P_X > P_Y > P_Z$

B $P_Z > P_Y > P_X$

C $P_X = P_Y = P_Z$

D $P_X > P_Z > P_Y$

E $P_Y > P_X > P_Z$

WORKED SOLUTION

Newton's second law gives cable tensions $m(g + a)$, mg and $m(g - a)$ at X , Y and Z , respectively. At each instant the cable force and velocity are parallel and the speed is the same, so $P = Tv$.

Therefore $P_X > P_Y > P_Z$, and the correct option is **A**.

Question 27

A nucleus initially has mass number A , proton number Z and neutron number $N = A - Z$. It emits some alpha particles and some beta-minus particles. Its final mass number is $A - 12$, and its final neutron number is $N - 8$. How many emissions of each type occurred, and what is the final proton number?

A 3 alpha, 2 beta-minus; $Z - 8$

B 3 alpha, 4 beta-minus; $Z - 2$

C 2 alpha, 2 beta-minus; $Z - 2$

D 3 alpha, 2 beta-minus; $Z - 6$

E 3 alpha, 2 beta-minus; $Z - 4$

WORKED SOLUTION

Only alpha decay changes mass number, by -4 each time. A change from A to $A - 12$ therefore requires three alpha decays. These reduce neutron number by 2 each, taking N to $N - 6$.

Each beta-minus decay converts a neutron into a proton, so it reduces neutron number by 1 . Reaching $N - 8$ therefore requires two beta-minus decays. The proton number changes by

$$-2(3) + 1(2) = -4,$$

so the final proton number is $Z - 4$. The correct option is **E**.

ANSWER KEY

ESAT 2024-25 Recall Mock - Chemistry

27 hard, recall-informed Chemistry questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Chemistry answer key

Q1	C	Q2	A	Q3	E	Q4	B	Q5	D	Q6	B	Q7	A	Q8	D	Q9	C
Q10	E	Q11	C	Q12	A	Q13	E	Q14	B	Q15	D	Q16	A	Q17	C	Q18	E
Q19	B	Q20	D	Q21	E	Q22	C	Q23	A	Q24	D	Q25	B	Q26	C	Q27	A

Chemistry

27 hard, recall-informed Chemistry questions with complete worked solutions.

C-01

Isotopes and probability

Answer **C**

Question 1

Element X has two naturally occurring isotopes, ^{63}X and ^{65}X . Its relative atomic mass is 63.5. Atoms combine at random to form X_2 molecules. What percentage of X_2 molecules are expected to have relative molecular mass 128?

A 12.5%**B** 25.0%**C** 37.5%**D** 50.0%**E** 75.0%

WORKED SOLUTION

Let the fraction of ^{65}X be p . Then $63(1-p) + 65p = 63.5$, so $p = 0.25$ and the ^{63}X fraction is 0.75. Mass 128 requires one atom of each isotope, in either order: $2(0.75)(0.25) = 0.375 = 37.5\%$.

Therefore the correct option is **C**.

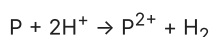
C-02

Ions, formulae and limiting amounts

Answer **A**

Question 2

Atoms P and Q have electron configurations P: 2,8,2 and Q: 2,7. In a two-stage process, 0.240 mol of P atoms and 0.360 mol of Q atoms first react as completely as possible. The ionic solid is removed, and any unreacted P is then added to excess acid:



Which option correctly gives the ionic solid formed in the first stage and the amount of hydrogen formed in the second?

A 0.180 mol of PQ_2 ; 0.060 mol H_2 **B** 0.240 mol of PQ_2 ; no H_2 **C** 0.120 mol of P_2Q ; 0.120 mol H_2 **D** 0.180 mol of covalent PQ_2 ; 0.060 mol H_2 **E** 0.360 mol of PQ_2 ; 0.120 mol H_2

WORKED SOLUTION

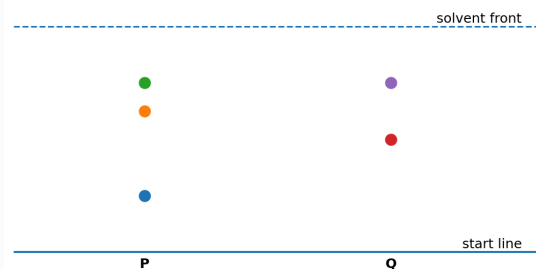
P forms P^{2+} and Q forms Q^- , so the ionic formula is PQ_2 . The 0.360 mol of Q atoms can form $0.360/2 = 0.180$ mol PQ_2 , using 0.180 mol P. This leaves $0.240 - 0.180 = 0.060$ mol P. The acid equation has a 1:1 ratio between P and H_2 , so the second stage forms 0.060 mol H_2 .

Therefore the correct option is **A**.

Distractor check: B ignores the 1:2 formula ratio and the leftover P; C reverses the ionic formula and doubles the gas; D has the right amounts but the wrong bonding type; E equates the amount of Q atoms with the amount of compound.

Question 3

The chromatograms P and Q were run with the same solvent and stationary phase. The solvent front travelled 8.0 cm in each case. In P, spot distances from the start are 2.0, 5.0 and 6.0 cm. In Q, they are 4.0 and 6.0 cm. Reference substances have R_f values W=0.25, X=0.50, Y=0.625 and Z=0.75. Which statement is correct?



- A** P contains W, X and Y
- B** P and Q both contain X
- C** Only Q contains Z
- D** Q contains W and Y
- E** P contains W, Y and Z, while Q contains X and Z

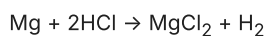
WORKED SOLUTION

For P, R_f values are 0.25, 0.625 and 0.75, so P contains W, Y and Z. For Q they are 0.50 and 0.75, so Q contains X and Z.

Therefore the correct option is **E**.

Question 4

2.40 g of magnesium is added to 100 cm³ of 1.00 mol dm⁻³ hydrochloric acid. When gas production stops, the unreacted magnesium is separated without loss and added to 100 cm³ of 0.500 mol dm⁻³ hydrochloric acid.



Take $A_r(\text{Mg})=24$ and molar gas volume at rtp as 24 dm³ mol⁻¹. What total volume of hydrogen is produced across the two stages?

A 0.60 dm³

B 1.80 dm³

C 2.40 dm³

D 1.20 dm³

E 3.60 dm³

WORKED SOLUTION

Initially there are $2.40/24=0.100$ mol Mg and 0.100 mol HCl. The first acid portion therefore forms 0.050 mol H₂ and consumes 0.050 mol Mg, leaving 0.050 mol Mg. The second acid portion contains $0.100 \times 0.500 = 0.050$ mol HCl, so it forms a further 0.025 mol H₂. The total is 0.075 mol, occupying $0.075 \times 24 = 1.80$ dm³ at rtp.

Therefore the correct option is **B**.

Distractor check: A counts only the second stage; C assumes all the original magnesium reacts; D counts only the first stage; E treats the HCl:H₂ ratio as 1:1 in both stages.

Question 5

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, the forward reaction is exothermic. Which single change increases both the equilibrium yield of ammonia and the initial forward reaction rate?

A decrease temperature

B increase temperature

C add a catalyst

D decrease the container volume

E remove the catalyst

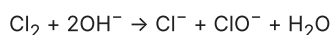
WORKED SOLUTION

Decreasing volume increases pressure, favouring the side with fewer gas molecules (the products), and also increases collision frequency. A catalyst increases rate but not equilibrium yield; lowering temperature improves yield but slows the reaction.

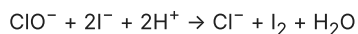
Therefore the correct option is **D**.

Question 6

In stage 1, 0.030 mol Cl_2 reacts completely with excess hydroxide ions:



The entire solution is then acidified and mixed with excess iodide ions. In stage 2, all the ClO^- reacts:



Which option gives the final amounts of Cl^- and I_2 , and the role of Cl_2 in stage 1?

- A** 0.030 mol Cl^- ; 0.030 mol I_2 ; both oxidising and reducing agent
- B** 0.060 mol Cl^- ; 0.030 mol I_2 ; both oxidising and reducing agent
- C** 0.060 mol Cl^- ; 0.060 mol I_2 ; oxidising agent only
- D** 0.030 mol Cl^- ; 0.060 mol I_2 ; reducing agent only
- E** 0.090 mol Cl^- ; 0.030 mol I_2 ; neither

WORKED SOLUTION

Stage 1 forms 0.030 mol Cl^- and 0.030 mol ClO^- . Stage 2 converts the 0.030 mol ClO^- into a further 0.030 mol Cl^- and forms 0.030 mol I_2 . The final chloride amount is therefore 0.060 mol. In stage 1, chlorine changes from oxidation state 0 to -1 in Cl^- and to +1 in ClO^- , so Cl_2 acts as both oxidising and reducing agent.

Therefore the correct option is **B**.

Distractor check: A omits the chloride made in stage 2; C doubles the iodine and ignores disproportionation; D reverses both the second-stage ratio and the redox role; E counts three chloride-producing routes although only two occur.

Question 7

Two pure solids are tested.

solid	melting point	electrical conduction	mechanical behaviour	water solubility
R	high	solid and molten	bends without shattering	insoluble
S	high	molten only	shatters when struck	soluble

Which explanation best fits all the evidence?

- A** R is metallic, with delocalised electrons; S is ionic, with ions mobile only when molten.
- B** R is ionic; S is metallic.
- C** R is graphite; S is ionic.
- D** R is metallic; S is giant covalent.
- E** R is simple molecular; S is ionic.

WORKED SOLUTION

R's high melting point, conduction in both states and ability to bend are consistent with a metal: delocalised electrons carry charge and layers of positive ions can slide while metallic bonding persists. S is consistent with an ionic solid: its ions are fixed in the solid but mobile in the melt; brittleness and water solubility provide supporting evidence.

Therefore the correct option is **A**.

Distractor check: B reverses the conductivity and mechanical patterns. C fits several observations but graphite is brittle rather than malleable. D fits R completely and some of S, but a typical giant covalent solid neither dissolves in water nor conducts when molten. E fits S but conflicts with R's high melting point and conduction.

Question 8

A halogen X_2 does not displace chloride ions from solution, but does displace bromide and iodide ions. Which statement is correct?

- A** X is bromine; X_2 is oxidised when it reacts with bromide ions.
- B** X is chlorine; bromine can oxidise X^- ions.
- C** X is fluorine; X_2 is reduced only when it reacts with iodide ions.
- D** X is chlorine; X_2 is reduced when it reacts with bromide or iodide ions.
- E** X is bromine; iodine can oxidise X^- ions.

WORKED SOLUTION

The displacement pattern identifies X as chlorine: chlorine does not displace chloride ions, but it is more reactive than bromine and iodine and so oxidises Br^- and I^- . In each displacement reaction, Cl_2 gains electrons to form Cl^- , so X_2 is reduced. Bromine cannot oxidise chloride ions because bromine is less reactive than chlorine.

Therefore the correct option is **D**.

Question 9

In each experiment, the same amount of calcium carbonate reacts with the same volume of excess hydrochloric acid.

experiment	acid concentration / mol dm ⁻³	temperature / °C	carbonate form	time to collect 40 cm ³ CO ₂ / s	final CO ₂ volume / cm ³
X	1.0	20	chips	80	48
Y	2.0	20	chips	45	48
Z	1.0	30	powder	20	48

Which conclusion is justified by the data?

- A** Y produces a greater final amount of carbon dioxide than X.
- B** Z proves that raising temperature alone makes the reaction faster.
- C** X and Y support the conclusion that higher acid concentration increases rate but not the final amount when carbonate is limiting.
- D** Equal final volumes show that all three reactions have the same rate.
- E** Z must contain the least calcium carbonate because it reaches 40 cm³ first.

WORKED SOLUTION

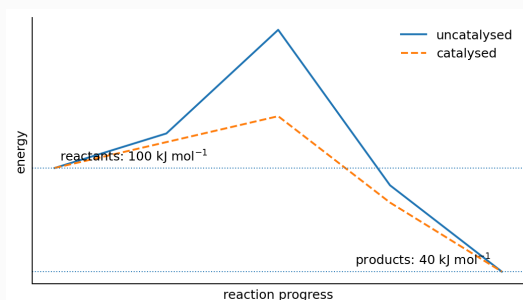
X and Y differ only in acid concentration. Y reaches 40 cm³ sooner, so it has a higher rate, while both reach 48 cm³ because the same amount of limiting calcium carbonate reacts. X and Z are not a controlled comparison for temperature because particle size also changes.

Therefore the correct option is **C**.

Distractor check: A contradicts the final volumes; B ignores the simultaneous surface-area change; D confuses amount with rate; E confuses reaching a volume sooner with producing less gas overall.

Question 10

The energy profile shows reactants at 100 kJ mol^{-1} , products at 40 kJ mol^{-1} , and a catalysed peak at 130 kJ mol^{-1} . What are ΔH for the forward reaction and the activation energy of the catalysed reverse reaction?

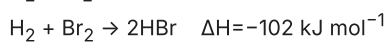
**A** -60 and 30 kJ mol^{-1} **B** $+60$ and 30 kJ mol^{-1} **C** $+60$ and 90 kJ mol^{-1} **D** -30 and 60 kJ mol^{-1} **E** -60 and 90 kJ mol^{-1} **WORKED SOLUTION**

$\Delta H = 40 - 100 = -60 \text{ kJ mol}^{-1}$. For the catalysed reverse reaction, $E_a = 130 - 40 = 90 \text{ kJ mol}^{-1}$.

Therefore the correct option is **E**.

Question 11

The following gaseous reactions have the stated enthalpy changes:



The average bond energies are $\text{H-H}=436$, $\text{Cl-Cl}=242$ and $\text{Br-Br}=194 \text{ kJ mol}^{-1}$. By how much is the average H-Cl bond energy greater than the average H-Br bond energy?

A 32.5 kJ mol^{-1}

B 41 kJ mol^{-1}

C 65 kJ mol^{-1}

D 82 kJ mol^{-1}

E 130 kJ mol^{-1}

WORKED SOLUTION

For HCl , $-184 = (436 + 242) - 2E(\text{H-Cl})$, so $E(\text{H-Cl}) = 862/2 = 431 \text{ kJ mol}^{-1}$. For HBr , $-102 = (436 + 194) - 2E(\text{H-Br})$, so $E(\text{H-Br}) = 732/2 = 366 \text{ kJ mol}^{-1}$. The difference is $431 - 366 = 65 \text{ kJ mol}^{-1}$.

Therefore the correct option is **C**.

Distractor check: A halves the final difference again; B uses only half the difference between the reaction enthalpies; D reports the reaction-enthalpy difference; E forgets that two H-X bonds form in each reaction.

Question 12

A strong acid has pH 2.00. A 10.0 cm^3 portion is diluted with water to 100.0 cm^3 . Then 20.0 cm^3 of this diluted acid is mixed with 180 cm^3 of $1.00\times 10^{-4}\text{ mol dm}^{-3}$ sodium hydroxide.

Assume complete dissociation and neutralisation, additive volumes and negligible ionisation of water. What is the final pH?

A 5.00**B** 3.00**C** 4.00**D** 6.00**E** 7.00**WORKED SOLUTION**

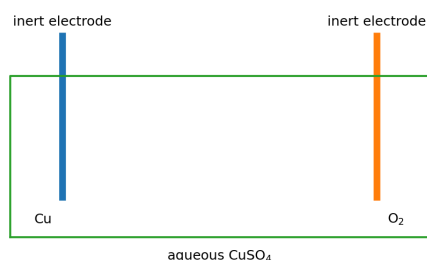
The original acid has $[\text{H}^+] = 1.00\times 10^{-2}\text{ mol dm}^{-3}$. Tenfold dilution makes this $1.00\times 10^{-3}\text{ mol dm}^{-3}$. The 20.0 cm^3 portion contains $2.00\times 10^{-5}\text{ mol H}^+$, while the alkali contains $0.180\times 1.00\times 10^{-4} = 1.80\times 10^{-5}\text{ mol OH}^-$. After neutralisation, $2.00\times 10^{-6}\text{ mol H}^+$ remains in 0.200 dm^3 , giving $[\text{H}^+] = 1.00\times 10^{-5}\text{ mol dm}^{-3}$ and pH 5.00.

Therefore the correct option is **A**.

Distractor check: B stops after the first dilution; C treats the mixing as dilution only and ignores hydroxide; D uses 2.00 dm^3 instead of 0.200 dm^3 as the final volume; E assumes exact neutralisation.

Question 13

Aqueous copper(II) sulfate is electrolysed with inert electrodes. Assume Cu^{2+} ions remain in excess throughout and that no other electrode reaction occurs. If 0.025 mol of oxygen is formed at the anode, what mass of copper is deposited at the cathode? Take $A_r(\text{Cu})=63.5$.

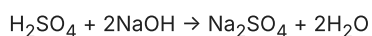
**A** 1.59 g**B** 6.35 g**C** 12.7 g**D** 25.4 g**E** 3.18 g**WORKED SOLUTION**

Forming 1 mol O_2 corresponds to 4 mol electrons, so 0.025 mol O_2 corresponds to 0.100 mol e^- . Cu^{2+} needs 2 e^- per Cu, so 0.050 mol Cu forms: $0.050 \times 63.5 = 3.175 \text{ g} \approx 3.18 \text{ g}$.

Therefore the correct option is **E**.

Question 14

25.0 cm³ of 0.200 mol dm⁻³ sulfuric acid is mixed with 40.0 cm³ of 0.250 mol dm⁻³ sodium hydroxide. Assume complete neutralisation and additive volumes.



A 13.0 cm³ aliquot of the resulting 65.0 cm³ mixture is removed. Then 5.0 cm³ of 0.200 mol dm⁻³ sodium hydroxide is added to what remains. Which statement describes the final mixture?

- A** It is neutral and contains 0.0040 mol of Na₂SO₄.
- B** It is alkaline and contains 0.0040 mol of Na₂SO₄.
- C** It is alkaline and contains 0.0050 mol of Na₂SO₄.
- D** It is neutral and contains 0.0050 mol of Na₂SO₄.
- E** It is acidic and contains 0.0010 mol of Na₂SO₄.

WORKED SOLUTION

Initially there are 0.00500 mol H₂SO₄ and 0.0100 mol NaOH, so they react exactly and form 0.00500 mol Na₂SO₄. Removing 13.0/65.0=1/5 of the homogeneous mixture removes 1/5 of the salt, leaving 0.00400 mol. The added sodium hydroxide is now in excess because no acid remains, so the final mixture is alkaline; it does not change the amount of Na₂SO₄.

Therefore the correct option is **B**.

Distractor check: A ignores the final sodium hydroxide; C ignores removal of the aliquot; D ignores both later operations; E reports the removed salt amount and reverses the acid–base conclusion.

Question 15

A 0.280 g sample of gaseous hydrocarbon X occupies 240 cm^3 at rtp. X decolourises bromine water. Take the molar gas volume at rtp as $24\text{ dm}^3\text{ mol}^{-1}$.

Which option gives the formula of X and the formula of the organic product when X reacts with bromine?

A CH_4 ; CH_3Br

B C_2H_6 ; $\text{C}_2\text{H}_6\text{Br}_2$

C C_2H_4 ; $\text{C}_2\text{H}_5\text{Br}$

D C_2H_4 ; $\text{C}_2\text{H}_4\text{Br}_2$

E C_3H_6 ; $\text{C}_3\text{H}_6\text{Br}_2$

WORKED SOLUTION

The gas amount is $0.240/24 = 0.010\text{ mol}$, so $M_r(\text{X}) = 0.280/0.010 = 28$. The hydrocarbon with M_r 28 is C_2H_4 . Its decolourisation of bromine water is consistent with a $\text{C}=\text{C}$ bond, and bromine adds across that bond to form $\text{C}_2\text{H}_4\text{Br}_2$.

Therefore the correct option is **D**.

Distractor check: C treats addition as substitution; E uses the formula of the next alkene without reconciling it with the measured molar mass.

Question 16

A 10.0 g sample of a metal carbonate is heated, but the decomposition is stopped before completion:



The heating produces $1.92\text{ dm}^3\text{ CO}_2$ at rtp. The cooled solid is then treated with excess acid, producing a further $0.48\text{ dm}^3\text{ CO}_2$ at rtp. In the acid, each mole of undecomposed MCO_3 produces one mole of CO_2 , while MO produces no gas. Take molar gas volume = $24\text{ dm}^3\text{ mol}^{-1}$, $A_r(\text{C}) = 12$ and $A_r(\text{O}) = 16$. What is $A_r(\text{M})$?

A 40

B 65

C 100

D 440

E 160

WORKED SOLUTION

Each mole of MCO_3 produces one mole of CO_2 , whether it decomposes during heating or reacts later with acid. The total gas volume is $1.92 + 0.48 = 2.40\text{ dm}^3$, corresponding to $2.40/24 = 0.100\text{ mol MCO}_3$. Therefore $M_r(\text{MCO}_3) = 10.0/0.100 = 100$, so $A_r(\text{M}) = 100 - (12 + 3 \times 16) = 40$.

Therefore the correct option is **A**.

Distractor check: B uses only the gas made during heating; C reports the relative formula mass of MCO_3 ; D uses only the undecomposed amount; E adds the carbonate-group mass instead of subtracting it.

Question 17

Two straight-chain compounds have condensed structural formulae $\text{CH}_2=\text{CHCH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}=\text{CHCH}_3$.

Which description is correct?

- A** They are the same compound written from opposite ends.
- B** They have different molecular formulae but belong to the same homologous series.
- C** They are structural isomers and both have molecular formula C_4H_8 .
- D** They have the same molecular formula but different functional groups.
- E** They are chain isomers and one is saturated.

WORKED SOLUTION

Both structures contain four carbon atoms, eight hydrogen atoms and one $\text{C}=\text{C}$ bond, so both have molecular formula C_4H_8 . The double bond is in a different position, so the displayed connectivity differs: they are structural (positional) isomers.

Therefore the correct option is **C**.

Distractor check: A confuses reversal of a formula with movement of the double bond; D overlooks that both compounds are alkenes.

Question 18

Metal M is below carbon in the reactivity series but above hydrogen. Consider the statements.

1. An oxide of M can be reduced by carbon under suitable conditions.
 2. M reacts with dilute hydrochloric acid to release hydrogen.
 3. During electrolysis of an aqueous salt of M using inert electrodes, M is deposited at the cathode in preference to hydrogen.
- Which statements are correct?

- A** 1 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 2 and 3
- E** 1 and 2 only

WORKED SOLUTION

Because M is below carbon, carbon can reduce its oxide under suitable conditions, so statement 1 is correct. Because M is above hydrogen, it can displace hydrogen from dilute acids, so statement 2 is correct. But for an aqueous electrolyte containing ions of a metal above hydrogen, hydrogen is discharged at the cathode in preference to the metal, so statement 3 is false.

Therefore the correct option is **E**.

Question 19

An aqueous salt gives a green precipitate with sodium hydroxide solution. A separate sample is acidified with dilute hydrochloric acid and then gives a white precipitate with barium chloride solution. Which inference is best?

- A** The salt contains Fe^{2+} and Cl^- ; acid is added to make the green precipitate form.
- B** The salt contains Fe^{2+} and SO_4^{2-} ; acidification prevents carbonate ions giving a misleading barium precipitate.
- C** The salt contains Cu^{2+} and SO_4^{2-} ; acid changes a blue precipitate to green.
- D** The salt contains Mg^{2+} and SO_4^{2-} ; barium chloride identifies the cation.
- E** The salt contains Fe^{2+} and NO_3^- ; hydrochloric acid supplies the sulfate ions.

WORKED SOLUTION

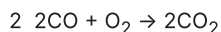
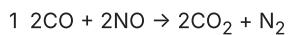
The green hydroxide precipitate indicates Fe^{2+} . A white precipitate after acidification and addition of barium chloride indicates sulfate ions. Acidification removes carbonate ions as carbon dioxide, preventing insoluble barium carbonate from causing a false positive.

Therefore the correct option is **B**.

Distractor check: A assigns the wrong anion and purpose to the acid; C and D mismatch the hydroxide colours; E wrongly claims hydrochloric acid contains sulfate.

Question 20

A 160 cm^3 exhaust pulse contains 90 cm^3 CO, 60 cm^3 NO and 10 cm^3 O₂. In a model catalytic converter, reaction 1 goes to completion before reaction 2 occurs:



All gas volumes are measured under the same conditions. Which option gives the final total gas volume and identifies the oxidising agent in each reaction?

A 90 cm^3 ; reaction 1: CO; reaction 2: CO

B 110 cm^3 ; reaction 1: NO; reaction 2: CO

C 120 cm^3 ; reaction 1: CO; reaction 2: NO

D 120 cm^3 ; reaction 1: NO; reaction 2: O₂

E 150 cm^3 ; reaction 1: O₂; reaction 2: O₂

WORKED SOLUTION

Reaction 1 consumes 60 cm^3 CO and all 60 cm^3 NO, forming 60 cm^3 CO₂ and 30 cm^3 N₂; 30 cm^3 CO remains. Reaction 2 uses all 10 cm^3 O₂ and 20 cm^3 of that CO, forming another 20 cm^3 CO₂. The final gases are 10 cm^3 CO, 80 cm^3 CO₂ and 30 cm^3 N₂, totalling 120 cm^3 . NO is reduced in reaction 1 and O₂ is reduced in reaction 2, so each is the oxidising agent in its reaction.

Therefore the correct option is **D**.

Distractor check: A treats the reducing agent as the oxidising agent in both stages; B identifies the first agent but loses a product volume and misclassifies CO in reaction 2; C finds the volume but reverses the redox roles; E ignores the first-stage reduction of NO and both gas contractions.

Question 21

A salt has solubility 80 g per 100 g water at 80°C and 20 g per 100 g water at 20°C. A 270 g saturated solution at 80°C is kept at 80°C while 30 g of water evaporates; any crystals formed remain in the vessel. The contents are then cooled to 20°C with no further water loss. What total mass of salt is present as crystals at 20°C?

A 24 g**B** 72 g**C** 90 g**D** 120 g**E** 96 g**WORKED SOLUTION**

At 80°C, 180 g saturated solution contains 100 g water and 80 g salt, so 270 g contains 150 g water and 120 g salt. Evaporation leaves 120 g water. At 80°C this can keep 96 g salt dissolved, so 24 g has already crystallised. At 20°C, 120 g water can hold only 24 g salt, so the total crystal mass is $120 - 24 = 96$ g.

Therefore the correct option is **E**.

Distractor check: A gives either the salt crystallised during evaporation or the salt left dissolved at 20°C; B counts only the additional crystallisation on cooling; C ignores the water loss; D assumes all salt crystallises.

Question 22

At the same temperature and pressure, 30 cm³ of gaseous A reacts exactly with 45 cm³ of gaseous B to form 30 cm³ of gaseous C only. Which equation has the simplest whole-number coefficients consistent with the data?

A $A + 1.5B \rightarrow C$ **B** $2A + 3B \rightarrow 3C$ **C** $2A + 3B \rightarrow 2C$ **D** $3A + 2B \rightarrow 3C$ **E** $4A + 6B \rightarrow 4C$ **WORKED SOLUTION**

At the same temperature and pressure, gas volumes are proportional to amounts in moles. The ratio 30:45:30 simplifies to 2:3:2. Therefore the simplest whole-number equation is $2A + 3B \rightarrow 2C$. Option E has the same ratio but its coefficients are not in simplest form.

Therefore the correct option is **C**.

Question 23

A student burns 0.010 mol of a fuel. The combustion raises the temperature of 200 g of water by 12.5°C. The specific heat capacity of water is 4.2 J g⁻¹ °C⁻¹.

Assume that 70% of the energy released by the fuel is transferred to the water. What is the molar energy release of the fuel?

A 1500 kJ mol⁻¹**B** 1050 kJ mol⁻¹**C** 735 kJ mol⁻¹**D** 150 kJ mol⁻¹**E** 15 kJ mol⁻¹**WORKED SOLUTION**

The water gains $q = mc\Delta T = 200 \times 4.2 \times 12.5 = 10\,500$ J. This is 70% of the fuel's energy release, so the fuel releases $10\,500 / 0.70 = 15\,000$ J for 0.010 mol. Per mole this is $15\,000 / 0.010 = 1\,500\,000$ J mol⁻¹ = 1500 kJ mol⁻¹.

Therefore the correct option is **A**.

Distractor check: B ignores the 70% transfer; C applies the efficiency in the wrong direction; D and E lose powers of ten during the per-mole or joule-to-kilojoule conversion.

Question 24

On complete combustion, assume that all sulfur in a fuel is converted to sulfur dioxide.

fuel	mass of sulfur burned	energy released
P	0.80 g	40 MJ
Q	0.30 g	10 MJ

Take $A_r(\text{S})=32$ and $A_r(\text{O})=16$. Which row gives the sulfur dioxide emitted per MJ and identifies the lower-emission fuel?

A P: 0.020 g MJ^{-1} ; Q: 0.030 g MJ^{-1} ; P is lower

B P: 0.040 g MJ^{-1} ; Q: 0.060 g MJ^{-1} ; Q is lower

C P: 0.080 g MJ^{-1} ; Q: 0.120 g MJ^{-1} ; P is lower

D P: 0.040 g MJ^{-1} ; Q: 0.060 g MJ^{-1} ; P is lower

E P: 0.040 g MJ^{-1} ; Q: 0.030 g MJ^{-1} ; Q is lower

WORKED SOLUTION

$M_r(\text{SO}_2)=64$, so each gram of sulfur forms $64/32=2 \text{ g}$ of SO_2 . Fuel P therefore emits $1.60/40=0.040 \text{ g MJ}^{-1}$; fuel Q emits $0.60/10=0.060 \text{ g MJ}^{-1}$. P has the lower emission per unit energy.

Therefore the correct option is **D**.

Distractor check: A compares sulfur rather than sulfur dioxide; B calculates both rates but reverses the comparison; C applies the mass factor twice.

Question 25

A mixture contains water, dissolved sodium chloride and insoluble sand. Which sequence can produce both pure water and dry sodium chloride as separate products?

- A** Filter the mixture, then evaporate the filtrate to dryness while allowing the water vapour to escape.
- B** Filter the mixture. Distil the filtrate and collect the water, stopping before the residue becomes dry. Evaporate the remaining concentrated solution and dry the sodium chloride crystals.
- C** Distil the original mixture and then dry its residue, without first separating the sand.
- D** Filter the mixture, crystallise some sodium chloride, and use the remaining mother liquor as the pure-water product.
- E** Centrifuge the mixture and decant the liquid as pure water, then dry the sediment as sodium chloride.

WORKED SOLUTION

Filtration removes the sand. Simple distillation produces pure water as the distillate. The remaining concentrated sodium chloride solution is then evaporated and the salt is dried.

Therefore the correct option is **B**.

Distractor check: A recovers salt but not water; C leaves sand mixed with the salt; D mistakes salt solution for pure water; E cannot remove dissolved sodium chloride by centrifugation.

Question 26

A 0.300 g compound contains only carbon, hydrogen and oxygen. Complete combustion produces 0.440 g CO_2 and 0.180 g H_2O . What is the empirical formula? Take A_r : C=12, H=1, O=16.

- A** CHO
- B** $\text{C}_2\text{H}_4\text{O}$
- C** CH_2O
- D** CH_4O
- E** $\text{C}_2\text{H}_2\text{O}$

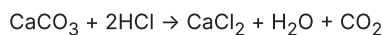
WORKED SOLUTION

$0.440 \text{ g CO}_2 = 0.010 \text{ mol CO}_2$, so the sample contained $0.010 \text{ mol C} = 0.120 \text{ g C}$. $0.180 \text{ g H}_2\text{O} = 0.010 \text{ mol H}_2\text{O}$, so it contained $0.020 \text{ mol H} = 0.020 \text{ g H}$. Oxygen mass is $0.300 - 0.140 = 0.160 \text{ g} = 0.010 \text{ mol O}$. Ratio C:H:O = 1:2:1, so CH_2O .

Therefore the correct option is **C**.

Question 27

A 10.0 g sample of impure calcium carbonate reacts with excess hydrochloric acid. A calibration shows that the gas-collection apparatus captures 75.0% of the carbon dioxide produced; the syringe reading is 1.26 dm³ at rtp. Assume that calcium carbonate is the only substance in the sample that produces carbon dioxide.



Take $M_r(\text{CaCO}_3)=100$ and molar gas volume = 24 dm³ mol⁻¹. What is the percentage purity?

A 70.0%**B** 52.5%**C** 35.0%**D** 93.3%**E** 28.0%**WORKED SOLUTION**

The actual carbon dioxide volume is $1.26/0.750=1.68$ dm³. This is $1.68/24=0.0700$ mol CO₂, so the sample contained 0.0700 mol CaCO₃, with mass $0.0700 \times 100 = 7.00$ g. The percentage purity is $(7.00/10.0) \times 100 = 70.0\%$.

Therefore the correct option is **A**.

Distractor check: B treats the captured volume as the full yield; C incorrectly applies the coefficient 2 from HCl; D corrects for the 75% capture twice; E uses $A_r(\text{Ca})=40$ instead of $M_r(\text{CaCO}_3)=100$.